

Emil Levo

Software Engineer
Computational Physicist

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Who Am I?

Software Engineer located in Helsinki, Finland. Currently working in the space sector, developing satellite ground processing software. My main areas of interest include programming and physics, but as long as I get to solve problems, I'm right at home.

Programming

C++ 
Python 
C# 
Bash 
Fortran 

Tools & Technologies

Linux, Docker, Git, GitLab CI/CD, CMake, Unity, Office

Experience

- 2024 – **Software Engineer** **Huld**
Satellite ground processing software development. Responsibilities include design and development, unit testing, validation, maintaining the development and CI/CD environments, and documentation.
C++ / Python / Docker / GitLab CI/CD / CMake / Office
- 2022 – 2023 **Game Programmer** **Heroic Games**
Various responsibilities related to game programming, including but not limited to, implementing AI, UI, various game mechanics, and shaders.
Unity / C# / Shader Graph / Quantum - Photon Engine
- 2019 – 2021 **Doctoral Researcher** **University of Helsinki**
Computational research of irradiation damage in high entropy alloys for fusion reactor applications. The work included running computationally intensive molecular dynamics irradiation simulations, analyzing irradiation damage in simulated samples, publishing results via peer-reviewed articles in journals related to the field, and teaching.
Molecular Dynamics / OVITO / Linux / Python
- 2015 – 2019 **Research Assistant** **University of Helsinki**
Computational research of irradiation damage in high entropy alloys for fusion reactor applications.
Molecular Dynamics / OVITO / Linux / Python

Education

- 2019 – 2022 **Doctoral Degree** **University of Helsinki**
Doctoral Programme in Materials Research and Nanoscience
- 2017 – 2019 **Master's Degree** **University of Helsinki**
Master's Programme in Materials Research
Line: Computational Material Physics
- 2014 – 2017 **Bachelor's Degree** **University of Helsinki**
Degree Programme in Physical Sciences
Major: Theoretical Physics

Teaching

- 2017 – 2019 **Teaching Assistant** **University of Helsinki**
Mathematics for Physicists I
Basics of vector calculus, power series, differential calculus, integral calculus

2017 – 2019	Teaching Assistant Mathematics for Physicists II Complex numbers, ordinary differential equations, vector calculus	University of Helsinki
2018 – 2021	Teaching Assistant Mathematics for Physicists III Linear algebra, vector calculus	University of Helsinki

International Experience

2019	ICFRM 2019 International conference on fusion reactor materials Poster presentation	San Diego
2014 – 2015	Lancaster University First year BSc physics studies.	Lancaster

Publications

- [1] Levo, E. (2022). Radiation Damage in High Entropy Alloys. University of Helsinki.
- [2] Levo, E., Granberg, F., Utt, D., Albe, K., Nordlund, K., & Djurabekova, F. (2019). Radiation stability of nanocrystalline single-phase multicomponent alloys. Journal of Materials Research.
- [3] Levo, E., Granberg, F., Fridlund, C., Nordlund, K., & Djurabekova, F. (2017). Radiation damage buildup and dislocation evolution in Ni and equiatomic multicomponent Ni-based alloys. Journal of Nuclear Materials, 490, 323-332.
- [4] Granberg, F., Djurabekova, F., Levo, E., & Nordlund, K. (2017). Damage buildup and edge dislocation mobility in equiatomic multicomponent alloys. Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms, 393, 114-117.

Languages

Swedish - native
Finnish - native
English - excellent

Hobbies

If not working, one can most certainly find me at the gym, playing games, cooking, developing passion projects, or planning my future garden.

Military Service

Served as a conscript in the Finnish Defence Forces for a year. Underwent non-commissioned officer training and gained valuable leadership skills.